

From Ranking to Retrieval: Purchase-Aware Tree Distillation for Recommendation

Naoyuki Tomita
X Pipelines Co., Ltd.
Ho Chi Minh City, Vietnam

ABSTRACT

Large-scale home feeds commonly separate candidate retrieval from a purchase-oriented ranker. This division creates an objective gap: a retriever is optimized from sparse interactions while the downstream ranker estimates a denser purchase utility. We formulate Purchase-aware Tree Distillation (PTD), a preregistered utility-projection method for a balanced item tree. A frozen Entire Space Multi-Task Model (ESMM) supplies item purchase log-odds at leaf decisions; for each internal child, PTD sums the purchase utility of eligible descendant items and normalizes the resulting masses across siblings. The resulting targets supervise the local decisions traversed by the tree, without an online teacher call. PTD also includes an HSTU-style action-sequence encoder and a train-only alternating tree update related to TDM/JTM. The evaluation contract fixes a point-in-time-safe evidence universe, a five-date temporal test, three random seeds, paired user bootstrap inference, and explicit quality, diversity, and latency guardrails. A verified legacy artifact establishes the common evidence universe and teacher anchor (12,844,800 test pairs; ESMM purchase NDCG@50 0.17626), but it contains no PTD result. An independent 24-cell RetailRocket validation is admitted as negative/null external-validity evidence: true-sum mass did not outperform max-descendant aggregation at the primary width. The prospective five-day Kauche tree-family results remain pending; consequently this draft makes no positive efficacy or priority claim. The accompanying artifact provides schemas, a claim-evidence ledger, deterministic table/figure generation, and fail-closed validation so that empirical claims can be promoted only after immutable manifests, hashes, and evaluation JSON are available.

CCS CONCEPTS

• **Information systems** → **Recommender systems; Retrieval models and ranking**; • **Computing methodologies** → *Knowledge representation and reasoning*.

KEYWORDS

candidate retrieval, recommender systems, tree-based retrieval, knowledge distillation, conversion modeling, sequential recommendation

1 INTRODUCTION

Industrial recommendation pipelines must retrieve a small candidate set before an expressive ranker can evaluate it. Two-tower retrieval makes this stage compatible with maximum-inner-product indexes, but constrains the interaction function. Tree-based deep models instead place items at leaves and evaluate user-node compatibility from root to leaf, supporting expressive scorers with logarithmic dependence on corpus size under fixed branching and beam assumptions [21]. Their supervision, however, is usually derived from observed positive paths and sampled negatives. The

downstream purchase ranker may encode a richer ordering than those binary path labels.

This paper studies a deliberately narrow question: can a fixed purchase model teach the local decisions made by a tree retriever? We call the registered method Purchase-aware Tree Distillation (PTD). The teacher is a frozen ESMM-style model whose purchase score factors as predicted click-through rate times predicted post-click conversion rate [6]. The teacher natively scores items in a flat candidate space, whereas the student must decide which child to traverse at each level. PTD bridges that structural mismatch by projecting item utility onto sibling-conditional targets: leaves retain purchase log-odds, and internal children receive the aggregate purchase mass of their eligible descendants. This supplies routing supervision before the target item is reached and avoids requiring a globally normalized teacher distribution over the complete catalog. The resulting mechanistic hypothesis is objective-conditional and falsifiable: max aggregation targets the best descendant, sum mass targets additive top- K purchase utility, and a finite beam may amplify their difference through irreversible early pruning.

Three components are integrated. First, an HSTU-style encoder represents the user’s point-in-time action sequence [15]. Second, a balanced tree constrains serving work and makes sibling groups explicit. Third, an optional JTM-style alternating phase updates item assignments from training evidence only and locks the tree before test scoring [20]. Figure 1 summarizes the design.

This is an evidence-bounded draft. We contribute (i) an applied formulation that combines purchase-oriented ranker supervision with tree-induced marginalization to train the internal and leaf decisions of an explicit candidate retriever, (ii) a preregistered temporal evaluation and ablation plan, and (iii) a reproducible artifact contract that separates verified, preregistered, and pending statements. Factorized and marginal structural distillation, ranker-to-retriever transfer, conversion-aware candidate generation, and beam-aware training all have precedents. We therefore make no “first” claim. The admitted RetailRocket result is negative/null, and we do not claim that PTD improves retrieval while the prospective Kauche evaluation remains pending.

2 PROBLEM FORMULATION

Let u denote a request with a history H_u available strictly before snapshot time t , and let I_t be eligible items. A balanced b -ary tree $\mathcal{T}$ maps each item i to a leaf. At depth h , beam search keeps B nodes according to a student score $s_\theta(u, n)$. The serving objective is to retrieve a set $R_K(u)$ with high future purchase utility while bounding the number of node evaluations.

Write the root-to-leaf code for item i as $\pi_{\mathcal{T}}(i) = (n_0, \dots, n_H)$ and the children competing with n_h as $S_h(i)$. With a balanced tree, $H = \lceil \log_b |I_t| \rceil$ up to capacity padding. The HSTU-style variant encodes the user once, after which beam retrieval evaluates at most

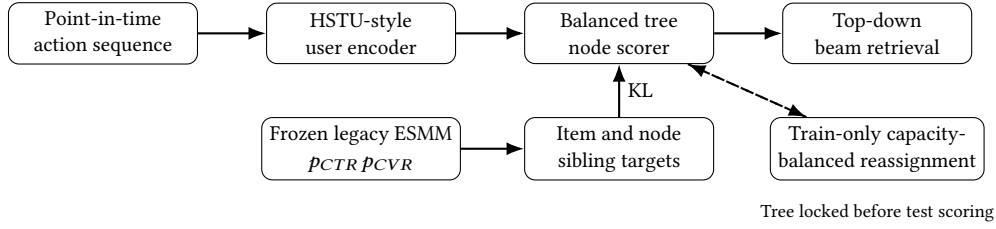


Figure 1: Registered PTD design. Solid arrows are model/data flow; dashed arrows occur only during train-side alternating optimization. The teacher is frozen.

$b + Bb(H - 1)$ child nodes before leaf truncation, or $O(BbH)$ scorer calls; the node-conditioned DIN ablation is timed separately. This bound motivates a local decision objective: the student need only order siblings encountered by the beam, rather than score every eligible item online.

For each eligible pair, the frozen teacher produces

$$q_T^{\text{purchase}}(u, i) = \hat{p}_T(\text{click} \mid u, i) \hat{p}_T(\text{purchase} \mid \text{click}, u, i), \quad (1)$$

following the ESMM factorization. This score is a training signal, not ground truth. It can inherit calibration, exposure, and representation biases from the legacy model.

The observed label y_{ui} is one only when a home-feed click-to-purchase event occurs in $[t, t + 24\text{h})$. All sequence, item, teacher-feature, and tree-construction inputs must precede t . Our primary metric is macro user purchase NDCG@50 over users with an observed purchase.

3 PURCHASE-AWARE TREE DISTILLATION

3.1 Point-in-time HSTU-style user encoder

The frozen input contains separate click and purchase product-ID streams, each most-recent-first and capped at 30. It does not retain per-event timestamps or a total order across streams. We reverse each stream independently to oldest-to-newest and apply the same two-layer, four-head, 64-dimensional gated sequential-transduction stack to both. Item and stream-type embeddings are shared, positional encoding is learned, and time encoding is disabled. Final non-padding click and purchase states are fused with a hashed-user state into z_u ; empty streams use a learned token. We use “HSTU-style” because the component adopts gated sequential transduction, not because it reproduces the scale or complete feature set of Zhai et al. [15]. A node representation and contextual features enter $s_\theta(z_u, e_n, x_{un})$.

The registered non-HSTU ablation consumes exactly the same two streams and embeddings through per-stream multiwindow DIN operators with windows (1, 2, 3, 4, 5, 5, 10). It shares the tree, scorer head, optimizer, examples, and width, but exact parameter equality is not claimed; trainable counts and measured latency are mandatory outputs. Thus RQ4 compares two fixed sequence operators without inventing unavailable temporal evidence.

3.2 Sibling teacher distributions

For a leaf-parent sibling set S_i , item-level teacher logits are

$$a_T(u, j) = \text{logit}(\text{clip}(q_T^{\text{purchase}}(u, j), \epsilon, 1 - \epsilon)), \quad j \in S_i. \quad (2)$$

For an internal child c , the teacher mass aggregates eligible descendant items from the training candidate universe,

$$m_T(u, c) = \sum_{j \in \text{desc}(c) \cap C_u} q_T^{\text{purchase}}(u, j), \quad (3)$$

and sibling logits are $a_T(u, c) = \log(m_T(u, c) + \epsilon)$. This aggregation is explicitly candidate-universe dependent; changing C_u is an experimental deviation.

Let $Y_{ui} \in \{0, 1\}$ be the purchase indicator and suppose the teacher is calibrated so that $q_T(u, i) = \mathbb{E}[Y_{ui} \mid u]$. Linearity of expectation gives

$$m_T(u, c) = \sum_{i \in \text{desc}(c) \cap C_u} q_T(u, i) = \mathbb{E} \left[\sum_{i \in \text{desc}(c) \cap C_u} Y_{ui} \mid u \right], \quad (4)$$

without requiring purchase indicators to be independent. Thus sum mass represents expected purchase count or additive utility recoverable from that subtree. The item scores need not sum to one across items, so this is generally not the probability of a mutually exclusive subtree event. In the special case of exactly one target and a teacher distribution normalized over all items, however, $m_T(u, c) = \Pr(i^* \in \text{desc}(c) \mid u)$ and the child conditional is exactly $m_T(u, c)/m_T(u, \text{parent}(c))$.

This interpretation also limits the claim. Max aggregation represents the best-one utility in a subtree, mean removes subtree cardinality from average utility, and log-sum-exp provides a smooth max-to-mass control. Sum is aligned with additive multi-item coverage, whereas max can be better for top-1 retrieval. Our primary scientific hypothesis is therefore not that sum always wins, but that appropriate subtree aggregation depends on the retrieval objective and that finite-beam pruning magnifies this objective mismatch.

The registered clipping constants are 10^{-6} for item logits and 10^{-12} for descendant mass. A child with no eligible descendant therefore receives finite, near-zero mass rather than being silently removed. Internal targets are recomputed only from the frozen training candidate universe. Multiplying every child mass in one sibling group by a common positive constant leaves its node target unchanged; the transfer is local, while the item target still reflects the teacher’s item-level log-odds.

At temperature τ , teacher and student sibling distributions are

$$Q_T(v | u, S) = \frac{\exp(a_T(u, v)/\tau)}{\sum_{w \in S} \exp(a_T(u, w)/\tau)}, \quad (5)$$

$$P_\theta(v | u, S) = \frac{\exp(s_\theta(u, v)/\tau)}{\sum_{w \in S} \exp(s_\theta(u, w)/\tau)}. \quad (6)$$

This is a registered hybrid projection, not a new general factorization theorem. Internal targets use additive subtree mass, but leaf targets preserve the ESMM teacher’s item-level log-odds. Consequently the local targets need not be marginals of one globally normalized structured teacher. Structural KD has already derived tractable factorized objectives, including incompatible teacher and student factorizations [11], and marginal distillation has summed descendant probabilities to reconcile hierarchical label spaces [1]. PTD’s narrower applied distinction is the use of downstream purchase utility to construct routing targets for an upstream explicit item-tree retriever.

For a purchase-positive training pair, the matched supervised path loss is

$$\mathcal{L}_{\text{tree}} = - \sum_{(u, i) \in D_{tr}: y_{ui}=1} \sum_{h=1}^H \log P_\theta(n_h | u, S_h(i)) \Big|_{\tau=1}. \quad (7)$$

The sibling denominator supplies path-local negatives. Every registered student uses Eq. 7; only the distillation terms, tree update, or encoder named by the ablation changes.

The complete objective is

$$\begin{aligned} \mathcal{L}(\theta; \mathcal{T}) = & \mathcal{L}_{\text{tree}} + \lambda_i \tau^2 \sum_{S_i} \text{KL}(Q_T^i \| P_\theta^i) \\ & + \lambda_n \tau^2 \sum_{h < S_H} \sum_{S_h} \text{KL}(Q_T^h \| P_\theta^h), \end{aligned} \quad (8)$$

The τ^2 factors preserve gradient scale under temperature smoothing. Item-only, node-only, combined, and no-distillation variants isolate the terms in Eq. 8. The teacher is evaluated without gradient and never updated from student or test data.

3.3 Train-only alternating optimization

The fixed tree is a deterministic binary depth-13 hierarchy over a preregistered 5,584-item catalog envelope, sorted by earliest point-in-time category and product ID. This uses test-period item identity/category but no label or teacher score; every date masks absent items before beam search. The design is therefore a disclosed transductive catalog comparison, not an inductive cold-item claim. The alternating variant follows JTM’s broad model/index co-optimization pattern but imposes a strict evidence boundary: only train-seen items may move, and only training interactions and frozen training teacher scores contribute assignment weights. Validation selects a cycle in $\{0, 1, 2, 3\}$ independently for each alternating variant and seed; the selected checkpoint, tree, and mask bundle is then immutable for test scoring.

For a candidate leaf ℓ with path $\pi(\ell)$, the registered train-only assignment weight is

$$W(i, \ell) = \sum_{u: (u, i) \in D_{tr}} \left(y_{ui} + q_T^{\text{purchase}}(u, i) \right) \sum_{h=1}^H \log P_\theta(\ell_h | u, S_h(\ell)) \Big|_{\tau=1}. \quad (9)$$

Algorithm 1 Registered PTD training and tree lock

Require: train D_{tr} , validation D_{va} , initial tree $\mathcal{T}_0$, frozen teacher T

- 1: Verify point-in-time and candidate-universe assertions
 - 2: **for** $r = 0, \dots, 3$ **do**
 - 3: Fit θ_r on D_{tr} for two epochs using Eq. 8
 - 4: if $r > 0$, warm-start parameters from θ_{r-1} ; reset optimizers
 - 5: Record validation NDCG@50 without changing T
 - 6: **if** $r < 3$ **then**
 - 7: Compute train-only assignment weights using Eq. 9
 - 8: $\mathcal{T}_{r+1} \leftarrow$ capacity-constrained reassignment
 - 9: **end if**
 - 10: **end for**
 - 11: Select validation maximum r^* ; exact ties choose lower r
 - 12: Lock $(\theta_{r^*}, \mathcal{T}_{r^*})$ and masks before test
 - 13: Score each test date once; emit manifest and evaluation JSON
-

The reassignment maximizes $\sum_i W(i, A(i))$ subject to one leaf per item and the declared subtree capacities. Equation 9 lets observed purchases and frozen soft utility weight the same model-path affinity; neither test labels nor test teacher scores enter it. The artifact includes a deterministic small-instance reference and an exact rectangular maximum-weight solver. The latter removes leaves occupied by anchored non-training items, matches every train-seen item to one remaining physical leaf, and verifies all capacity constraints.

This procedure is not equivalent to end-to-end differentiable tree learning. It is also not teacher-free JTM: the fixed purchase model influences local student targets and, in the alternating variant, train-side assignment weights.

3.4 Serving and training cost

Teacher inference, descendant aggregation, and capacity-constrained reassignment are offline operations. At retrieval, each binary sibling set is normalized after masking children with no date-eligible descendant; root-to-leaf log probabilities accumulate before deterministic node-ID tie-breaking and beam truncation. Internal and padding nodes use distinct hash keys, while leaves retain their locked product and category representation. The HSTU-style path caches one user state per query; the DIN ablation remains node-conditioned. Distillation adds neither a teacher call nor a second-stage reranker to these deployable variants. Local distillation is aligned with the unit of a traversal decision, but it does not directly optimize the final set returned by a finite beam: an early prefix can still be pruned irreversibly. We therefore treat beam mismatch as a residual empirical question, using a fixed traversal protocol and an OTM baseline, rather than claiming to solve it theoretically. Latency is measured rather than inferred because batching, encoder cost, and hardware kernels can dominate node-count differences.

4 EXPERIMENTAL DESIGN

4.1 Evidence universe and splits

The verified source manifest fixes training to 2026-07-18–20, validation to 2026-07-21, and test to five dates from 2026-08-12 through 2026-08-28. The test spans 16 days. Every feature ends before its

Table 1: Verified legacy context on the common five-date universe. These are not PTD results.

Quantity	Verified value
Test rows	12,844,800
Purchase-positive pairs	1,136
Users with purchase	1,109
Legacy ESMM NDCG@50	0.17626
Legacy ESMM Recall@50	0.42065
Legacy ESMM AUC	0.79380

snapshot and labels occupy the next 24 hours. A metadata-only inventory pins 216 raw Parquet generations (23,761,140 rows), verifies identical split totals, and exposes the sequence, category, and feature-cutoff fields needed by PTD. A second outcome-free audit finds 3,126 train-union and 4,592 test-union items, including 2,339 absent from train and validation; it fixes the 5,584-item envelope and date masks by hash without publishing identifiers. These fields may join only on fixed row keys and cannot change eligibility or candidate counts.

Table 1 is the admitted internal empirical anchor. The source manifest and evaluation JSON were independently retrieved; their SHA-256 hashes are recorded in the artifact. The table establishes a teacher/context anchor, not a PTD comparison.

An independent Category C validation on RetailRocket supplies a completed PTD aggregation comparison without changing the pending Kauche study. Across 116 validation users with transactions among 65,624 targets (evaluable fraction 0.001768), three seeds, and search widths 300–2400, the preregistered primary contrast (true-sum mass minus max-descendant transaction Recall@600 at width 1200) was -0.0231 (95% bootstrap CI $[-0.0526, 0.00361]$; Holm-adjusted $p = 0.378$). The interval’s upper endpoint is 0.00361, so effects larger than 0.36 Recall percentage points are outside this registered interval; this is not an equivalence conclusion. Thus the public-data result is admissible negative/null evidence: it supports neither true-sum superiority nor the registered beam-gap-shrinkage mechanism. Six width-4800 attempts produced no immutable cell artifact and remain recorded as non-evidence; the confirmatory claim is explicitly limited to width 2400. The locked test split remains unopened.

The descriptive width effects are -0.0033 , -0.0216 , -0.0231 , and -0.0408 at widths 300, 600, 1200, and 2400. All three seed estimates are negative at widths 1200 and 2400; the smaller widths show mixed seed signs. Mean candidate-set Jaccard is only 0.151–0.201, so the result is not explained by the two aggregators converging to the same candidates. Every query first prunes at depth 11, 12, 13, and 14 as width increases; the first-prune retained/evaluated ratio remains about 0.293. Relative to width 300, completed wider cells use 45–71% higher mean p95 latency and 52–104% higher mean peak GPU memory, without a true-sum advantage. These are descriptive diagnostics, not additional hypothesis tests.

4.2 Baselines, ablations, and metrics

The registered family contains fixed-tree TDM-style supervision, item-only PTD, node-only PTD, combined PTD, train-only alternating supervision, alternating PTD, and a matched encoder ablation. We also retain an ESMM reranking oracle over the same candidate universe as a diagnostic upper anchor. TDM [21], JTM [20], beam-calibrated OTM [22], and same-level competition in DTR/TDR [5] motivate distinct baseline dimensions rather than interchangeable names.

The primary outcome is purchase NDCG@50 averaged over seeds 16630–16632. Secondary outcomes include Recall@50, NDCG@10/100, purchase AUC, click NDCG@50, diversity, candidate count, and p50/p95 latency. Date-stratified paired user bootstrap intervals use 10,000 resamples with seed 20260925. Four primary contrasts use Holm correction; the RQ2 comparator is the better single-level variant chosen on validation seed 16630 and locked before test. Every candidate’s guardrails use fixed-tree TDM: quality and diversity allow at most 5% relative degradation, and p95 latency at most 1.20 times baseline under matched hardware, single concurrency, 100 warm-up, and 1,000 measured queries. This relative threshold is not a production SLO.

4.3 Estimand and statistical procedure

For a date–user unit (d, u) , let M_{dus}^A and M_{dus}^B be purchase NDCG@50 for variants A and B under seed s . The paired seed-averaged effect is

$$\delta_{du}^{A,B} = \frac{1}{3} \sum_{s \in \{16630, 16631, 16632\}} (M_{dus}^A - M_{dus}^B), \quad \hat{\Delta}^{A,B} = \frac{1}{N} \sum_{d,u} \delta_{du}^{A,B}. \quad (10)$$

Only users with an observed purchase for the macro purchase metric enter N ; a user appearing on two dates contributes two date–user units. Each bootstrap replicate samples date–user units with replacement within each date, preserving that date’s original unit count, and then recomputes Eq. 10. The 2.5th and 97.5th percentiles form the interval. A two-sided sign probability yields the raw p -value; Holm’s step-down procedure adjusts the four RQ contrasts. Success requires a positive adjusted conclusion, a confidence-interval lower bound above zero, and every registered guardrail to pass. Secondary metrics and all slice analyses remain descriptive.

Purchase and click NDCG use binary relevance, logarithmic discount, and an ideal ranking over all eligible items; a zero-positive purchase query scores zero and is excluded only from the registered purchase-positive macro average. Recall uses the complete eligible positive count. Purchase AUC uses retrieval rank over all eligible items, with unretrieved items tied below top- K and single-class queries set to 0.5. Category coverage@50 is distinct retrieved categories divided by $\min(50, \text{eligible distinct categories})$; maximum share is the largest top-50 category frequency divided by returned length. Latency excludes data loading and teacher computation but includes synchronized user encoding, traversal, and top- K materialization after at least 100 serial warm-ups and over at least 1,000 measured queries per variant.

Table 2: RetailRocket Category C results (116 purchase users / 65,624 target users; evaluable fraction 0.001768). Intervals by width are descriptive applications of the registered bootstrap; only the width-1200 primary contrast belongs to the registered Holm family ($p_{\text{Holm}} = 0.378$).

Width	Δ Recall@600	descriptive 95% CI	mean Jaccard	first prune	p95 ms	peak GiB
300	-0.0033	[-0.0381, 0.0306]	0.151	11	18294	8.5
600	-0.0216	[-0.0549, 0.0085]	0.190	12	31303	17.2
1200	-0.0231	[-0.0526, 0.0036]	0.201	13	29306	13.3
2400	-0.0408	[-0.0739, -0.0103]	0.195	14	26576	12.9

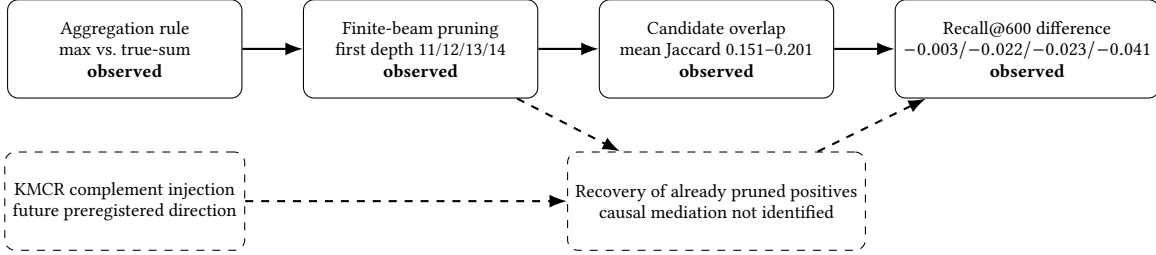


Figure 2: Descriptive failure-mechanism boundary. Solid boxes/arrows summarize observed 24-cell artifacts; dashed paths are unverified causal explanations or future work. The low overlap shows that the null/negative result is not candidate-set convergence, while wider search delayed first pruning without producing a true-sum advantage.

Table 3: Preregistered prospective comparisons. Values remain pending until the evidence gate passes.

Variant	NDCG@50	Δ [95% CI]	Status
Fixed-tree TDM	-	-	pending
PTD item sibling	-	-	pending
PTD node sibling	-	-	pending
PTD item+node	-	-	pending
Alternating TDM	-	-	pending
Alternating PTD	-	-	pending
PTD baseline encoder	-	-	pending
ESMM reranking oracle	-	-	pending

The dashes in Table 3 are intentional. One-day diagnostics cannot populate it; values require a schema-valid five-day evaluation, an immutable run manifest, and hash-linked date–user–seed records that reproduce every primary contrast.

4.4 Registered diagnostics and failure analysis

Every variant must report all seeds and dates even when it loses. The artifact records per-date metrics, cold-start cohorts, category concentration, candidates scored, and latency. Diagnostics may explain defects or failed guardrails but cannot change the primary variant, date, seed, cutoff, or tree. Missing seeds, non-finite scores, duplicate keys, universe mismatch, unlocked trees, or post-test deviations block admission. Valid null results and guardrail failures are admitted and reported but cannot satisfy positive claims; no failed slice may be deleted.

The completed public analysis narrows the supported boundary. True-sum is theoretically information-preserving for additive descendant utility, but changing internal aggregation cannot restore an item after its branch has been pruned. On RetailRocket, widening search delayed first pruning by three levels and produced materially different candidate sets, yet did not yield a true-sum Recall@600 advantage. Compact traces establish pruning locations

and counts, not causal mediation through individual positive-item survival; that causal path remains unverified. Descriptive purchase-positive-count slices are retained in the artifact but cannot create a post-hoc efficacy claim. Kauche historical results remain bounded context and are never pooled with Category C.

Future work shifts toward preregistered KMCR complement injection, which can add candidates absent from the base pre-union set. KMCR is a future direction only: its generator, complement budget, union order, tie-breaking, seeds, and stopping rules must be frozen before outcome access, and canary success is not efficacy evidence. No width-4800 rerun or new PTD superiority search is part of this paper update.

4.5 Reproducibility and claim promotion

The artifact fixes schemas for run, evaluation, and paired rows; pre-registration and amendments; a claim ledger; deterministic generators; and fail-closed validators. The checker requires all variants, seeds, dates, hashes, identities, and pre-test tree locks, then recomputes primary summaries, four bootstraps, Holm values, unit counts, and guardrail flags. Mismatch or deviation blocks admission. Admission does not imply efficacy: wording follows the verified sign, interval, adjusted p -value, and guardrails. Executable specifications and synthetic tests cover the method and statistics; SMOKE_ONLY records cannot support empirical claims. Figure 3 shows the path; a claim stays pending if an evidence edge is absent.

5 RELATED WORK

TDM introduced expressive top-down tree retrieval for recommendation and considered tree learning [21]. JTM formalized alternating optimization of index and preference model with hierarchical user representations [20]. OTM analyzed mismatch between node-wise training and beam-search inference [22], while more

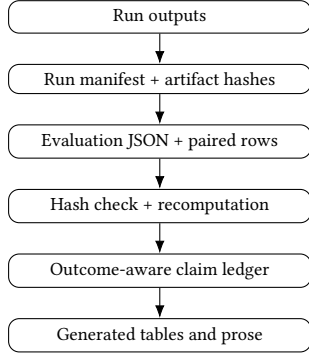


Figure 3: Evidence admission and interpretation. Invalid or irreproducible bundles stop at the gate; valid negative outcomes enter the ledger without supporting positive claims.

Table 4: Claim status at manuscript generation time.

Status	Claims
Verified	18
Preregistered	3
Pending	10

recent DTR/TDR work emphasizes same-level multi-class competition and sampled softmax [5]. OTM’s relevance-oriented pseudo-target and PTD’s additive expected purchase-utility mass are different target constructions, but tree retrieval, horizontal competition, and finite-beam mismatch are prior art.

Knowledge distillation transfers behavior from a teacher into a deployable student [2]. Ranking-specific work studies listwise or top-ranked transfer [7, 9], and Structural KD derives tractable local objectives when structured teacher and student factorizations agree, differ in granularity, or are incompatible [11]. In text retrieval, RocketQAv2 dynamically couples a dense retriever and reranker with listwise distillation [8]. Thus neither factorized KD nor ranker-to-retriever transfer is itself a PTD novelty.

Recommendation work has already established that independently trained nominators and rankers can interact poorly [3] and has transferred richer downstream objectives into candidate generation. DMTL distills click and reading-duration knowledge from a multi-task teacher into a candidate generator [18]; DMMP similarly transfers CTR/CVR knowledge into a multi-tower matching student [13]. CoRR aligns recommendation retriever and ranker distributions through cooperative training and KL distillation [4], while Full Stage Learning to Rank propagates downstream selection preferences across a multi-stage system [19]. A multi-objective content-recall patent also describes distilling click/conversion outputs into an efficient recall model [10]. These works preclude claims that PTD is the first purchase-aware, conversion-aware, or downstream-aligned retriever distillation method.

Recent generative recommenders narrow the boundary further. OneMall uses an online ranker as a multi-objective reward for generative retrieval [16]; BEAR and APAO explicitly optimize prefix

survival under beam search [12, 14]; SmartGR introduces hierarchy-aware and beam-aware distillation over semantic-ID generation [17]. PTD therefore does not claim the first hierarchical or beam-aware recommendation KD. Its registered distinction is the complete applied setting: item-level purchase utilities from a frozen downstream ranker are projected, through tree-induced descendant aggregation, onto sibling decisions of an upstream explicit tree retriever and evaluated under its serving-time finite-beam budget. ESMM supplies the frozen $pCTR \times pCVR$ score [6], and HSTU supplies an encoder component [15]; PTD modifies neither model.

6 LIMITATIONS AND THREATS TO VALIDITY

Construct validity. The ESMM score is a model-based proxy for purchase utility, not ground truth, user satisfaction, or welfare. It reflects historical exposure and can reproduce calibration, popularity, and representation biases. Internal-node mass is an expected utility/count mass over the frozen candidate universe, not a catalog-wide subtree probability. Moreover, leaf log-odds and internal additive mass form a hybrid local construction, not one globally consistent teacher factorization. Sum also has a concrete size-bias failure mode: a 100-item subtree with $q_i = 0.01$ has mass 1.0 and outranks a one-item subtree with $q_i = 0.9$, although max correctly prefers the latter for a best-one objective. Unequal subtree capacity, diffuse or duplicated utility, teacher miscalibration, NDCG discounting, and beam width one can produce related failures. The original registered ablations do not identify these mechanisms; causal attribution remains pending on the separate extension. Purchase NDCG emphasizes short-horizon conversion; the click, diversity, and concentration guardrails reduce but do not eliminate this mismatch.

Internal validity. The five test dates were used previously for legacy model evaluation, so the PTD analysis is prospective only with respect to the PTD readout, not blind to legacy outcomes. The tree also uses a fixed transductive envelope of test-period item identities and categories so that later items are representable; it never uses their labels or teacher scores, and this setup does not establish inductive new-item performance. Point-in-time assertions, fixed row keys, a frozen teacher, validation-only selection, immutable hashes, date masks, and a pre-test tree lock address identifiable leakage paths. They cannot rule out defects in upstream logging or unrecorded feature generation; a hash proves identity, not semantic correctness.

Statistical conclusion validity. Purchases are sparse: only 116 of 65,624 RetailRocket targets are purchase-evaluable. Pairing shared requests and seeds reduces variance but does not add independent populations. Holm correction covers only the four registered primary contrasts. Width-specific intervals beyond the registered primary contrast, candidate-overlap summaries, first-prune diagnostics, purchase-positive-count slices, secondary outcomes, and the teacher oracle are descriptive; absence of significance is not evidence of equivalence.

External validity. The completed aggregation comparison uses RetailRocket alone through width 2400; all six width-4800 attempts are terminal non-evidence. One catalog, one historical teacher, a

Table 5: Positioning against primary adjacent work. PTD statements describe the registered design, not empirical superiority.

Work	Prior mechanism	Registered PTD distinction
TDM / JTM [20, 21]	tree node / item leaf; teacher: none; tree: optional / alternating.	PTD retains coarse-to-fine retrieval and train-side reassignment but adds purchase-aware sibling targets from a frozen external ranker.
OTM / DTR [5, 22]	beam or same-level competing nodes; teacher: pseudo-target / none; tree: varies.	They address beam or horizontal competition; PTD transfers downstream purchase utility and treats finite-beam mismatch as residual.
Structural KD [11]	structured output factors; teacher: structured predictor; tree: none.	Factorized KD and incompatible factorizations are prior art; PTD instantiates marginalization between heterogeneous funnel stages.
Funnel transfer [4, 10, 13, 18, 19]	candidate items or funnel stages; teacher: engagement, CTR-CVR, or downstream ranker; tree: none.	Ranker-to-retriever and conversion-aware alignment are prior art; PTD instantiates them at explicit tree-routing decisions.
OneMall / SmartGR [12, 14, 16, 17]	generated item or semantic-ID prefix; teacher: ranker reward / large generator; tree: semantic tokenizer.	Purchase rewards and hierarchy/beam-aware objectives are prior art; PTD evaluates tree-induced purchase-utility targets in an explicit index.

fixed tree family, and one beam/latency protocol limit generalization. Local sibling KL does not guarantee the globally best top- K set under finite-beam pruning. The analysis does not prove equivalence, and its compact traces do not identify causal recovery of individual pruned positives. Offline Recall establishes neither online causal lift nor transfer to Kauche, other catalogs, or other hardware. Kauche evidence remains separately governed bounded context; a production claim would require a separately governed online experiment.

7 ETHICS AND SOCIETAL CONSIDERATIONS

Purchase optimization can over-emphasize high-propensity users or products, amplify exposure inequality, and encourage consumption. We record concentration and cold-start cohorts and forbid removal of failing slices. Private paired evidence uses access-controlled pseudonyms; the public artifact contains no row behavior. Review artifacts must be anonymous and policy compliant. Deployment requires separate safety and rollback review; an offline decision never changes production.

8 CONCLUSION

We specified Purchase-aware Tree Distillation as an applied combination of purchase-oriented ranker supervision and tree-induced marginalization for the sibling decisions of an explicit candidate retriever. The method uses additive expected descendant utility to create internal routing targets, while retaining teacher log-odds at leaves. The contribution is this heterogeneous funnel instantiation and its finite-beam evidence boundary, not factorized or marginal KD, downstream-objective alignment, hierarchy-aware KD, or beam-aware training in general. The admitted RetailRocket evidence shows a concrete boundary condition: an information-preserving internal aggregation still cannot recover candidates after finite-beam pruning, and delaying first prune through width 2400 did not establish true-sum superiority. This is a negative/null result, not an equivalence proof or a causal identification of the failure path. The artifact preserves every completed cell and every failed or cancelled attempt in a machine-checkable index. KMCR

complement injection is future preregistered work, not an efficacy claim. **[Pending: The separately governed five-day Kauche PTD effect sizes, confidence intervals, latency measurements, and ablation conclusions remain pending.]**

ACKNOWLEDGMENTS

This work was initiated during a one-month research visit to Fudan University.

REFERENCES

- [1] ASAPP, Inc. 2022. System, Method, and Computer Program for Obtaining a Unified Named Entity Recognition Model Using Marginal Distillation. U.S. Patent 11,487,944.
- [2] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. *arXiv preprint arXiv:1503.02531* (2015).
- [3] Jiri Hron, Karl Krauth, Michael I. Jordan, and Niki Kilbertus. 2021. On Component Interactions in Two-Stage Recommender Systems. In *Advances in Neural Information Processing Systems*, Vol. 34.
- [4] Xu Huang, Defu Lian, Jin Chen, Zheng Liu, Xing Xie, and Enhong Chen. 2023. Cooperative Retriever and Ranker in Deep Recommenders. In *Proceedings of the ACM Web Conference 2023*. 1150–1161. <https://doi.org/10.1145/3543507.3583422>
- [5] Ze Liu, Jin Zhang, Chao Feng, Defu Lian, Jie Wang, and Enhong Chen. 2024. Learning Deep Tree-based Retriever for Efficient Recommendation: Theory and Method. *arXiv preprint arXiv:2408.11345* (2024).
- [6] Xiao Ma, Liqin Zhao, Guan Huang, Zhi Wang, Zelin Hu, Xiaoqiang Zhu, and Kun Gai. 2018. Entire Space Multi-Task Model: An Effective Approach for Estimating Post-Click Conversion Rate. In *Proceedings of the 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1137–1140. <https://doi.org/10.1145/3209978.3210104>
- [7] Sashank Reddi, Rama Kumar Pasumarthi, Aditya Menon, Ankit Singh Rawat, Felix Yu, Seungyeon Kim, Andreas Veit, and Sanjiv Kumar. 2021. RankDistil: Knowledge Distillation for Ranking. In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (Proceedings of Machine Learning Research, Vol. 130)*. 2368–2376.
- [8] Ruiyang Ren, Yingqi Qu, Jing Liu, Wayne Xin Zhao, Qiaoqiao She, Hua Wu, Haifeng Wang, and Ji-Rong Wen. 2021. RocketQv2: A Joint Training Method for Dense Passage Retrieval and Passage Re-ranking. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. 2825–2835. <https://doi.org/10.18653/v1/2021.emnlp-main.224>
- [9] Jiaxi Tang and Ke Wang. 2018. Ranking Distillation: Learning Compact Ranking Models with High Performance for Recommender System. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2289–2298. <https://doi.org/10.1145/3219819.3220021>
- [10] Tencent Technology (Shenzhen) Co., Ltd. 2025. Content Recall Model Training Method, Content Recall Method, Device and Equipment. Chinese Patent CN113515690B.

- [11] Xinyu Wang, Yong Jiang, Zhaohui Yan, Zixia Jia, Nguyen Bach, Tao Wang, Zhongqiang Huang, Fei Huang, and Kewei Tu. 2021. Structural Knowledge Distillation: Tractably Distilling Information for Structured Predictor. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*. 550–564. <https://doi.org/10.18653/v1/2021.acl-long.46>
- [12] Weiqin Yang, Bohao Wang, Zhenxiang Xu, Jiawei Chen, Shengjia Zhang, Jingbang Chen, Canghong Jin, and Can Wang. 2026. BEAR: Towards Beam-Search-Aware Optimization for Recommendation with Large Language Models. *arXiv preprint arXiv:2601.22925* (2026).
- [13] Qingqing Yi, Jingjing Tang, Yifan Zeng, Xueting Zhang, and Weiqi Xu. 2024. DMMP: A Distillation-Based Multi-Task Multi-Tower Learning Model for Personalized Recommendation. *Knowledge-Based Systems* 284 (2024), 111236. <https://doi.org/10.1016/j.knosys.2023.111236>
- [14] Yuanqing Yu, Yifan Wang, Weizhi Ma, Zhiqiang Guo, and Min Zhang. 2026. APAO: Adaptive Prefix-Aware Optimization for Generative Recommendation. *arXiv preprint arXiv:2603.02730* (2026).
- [15] Jiaqi Zhai, Lucy Liao, Xing Liu, Yueming Wang, Rui Li, Xuan Cao, Leon Gao, Zhaojie Gong, Fangda Gu, Jiayuan He, Yinghai Lu, and Yu Shi. 2024. Actions Speak Louder than Words: Trillion-Parameter Sequential Transducers for Generative Recommendations. In *Proceedings of the 41st International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 235)*. 58484–58509.
- [16] Kun Zhang, Jingming Zhang, Wei Cheng, et al. 2026. OneMall: One Architecture, More Scenarios—End-to-End Generative Recommender Family at Kuaishou E-Commerce. *arXiv preprint arXiv:2601.21770* (2026).
- [17] Ziheng Zhang, Yu Cui, Bohao Wang, Yong He, Chao Yu, Chuan Yuan, Wujie Sun, Can Wang, and Jiawei Chen. 2026. SmartGR: Hierarchy and Beam-Aware Knowledge Distillation for Generative Recommendation. *arXiv preprint arXiv:2608.02048* (2026).
- [18] Zhong Zhao, Yanmei Fu, Hanming Liang, Li Ma, Guangyao Zhao, and Hongwei Jiang. 2021. Distillation based Multi-task Learning: A Candidate Generation Model for Improving Reading Duration. *arXiv preprint arXiv:2102.07142* (2021).
- [19] Kai Zheng, Haijun Zhao, Rui Huang, Beichuan Zhang, Na Mou, Yanan Niu, Yang Song, Hongning Wang, and Kun Gai. 2024. Full Stage Learning to Rank: A Unified Framework for Multi-Stage Systems. In *Proceedings of the ACM Web Conference 2024*. 3621–3631. <https://doi.org/10.1145/3589334.3645523>
- [20] Han Zhu, Daqing Chang, Ziru Xu, Pengye Zhang, Xiang Li, Jie He, Han Li, Jian Xu, and Kun Gai. 2019. Joint Optimization of Tree-based Index and Deep Model for Recommender Systems. In *Advances in Neural Information Processing Systems*, Vol. 32.
- [21] Han Zhu, Xiang Li, Pengye Zhang, Guozheng Li, Jie He, Han Li, and Kun Gai. 2018. Learning Tree-based Deep Model for Recommender Systems. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 1079–1088. <https://doi.org/10.1145/3219819.3219826>
- [22] Jingwei Zhuo, Ziru Xu, Wei Dai, Han Zhu, Han Li, Jian Xu, and Kun Gai. 2020. Learning Optimal Tree Models under Beam Search. *arXiv preprint arXiv:2006.15408* (2020).